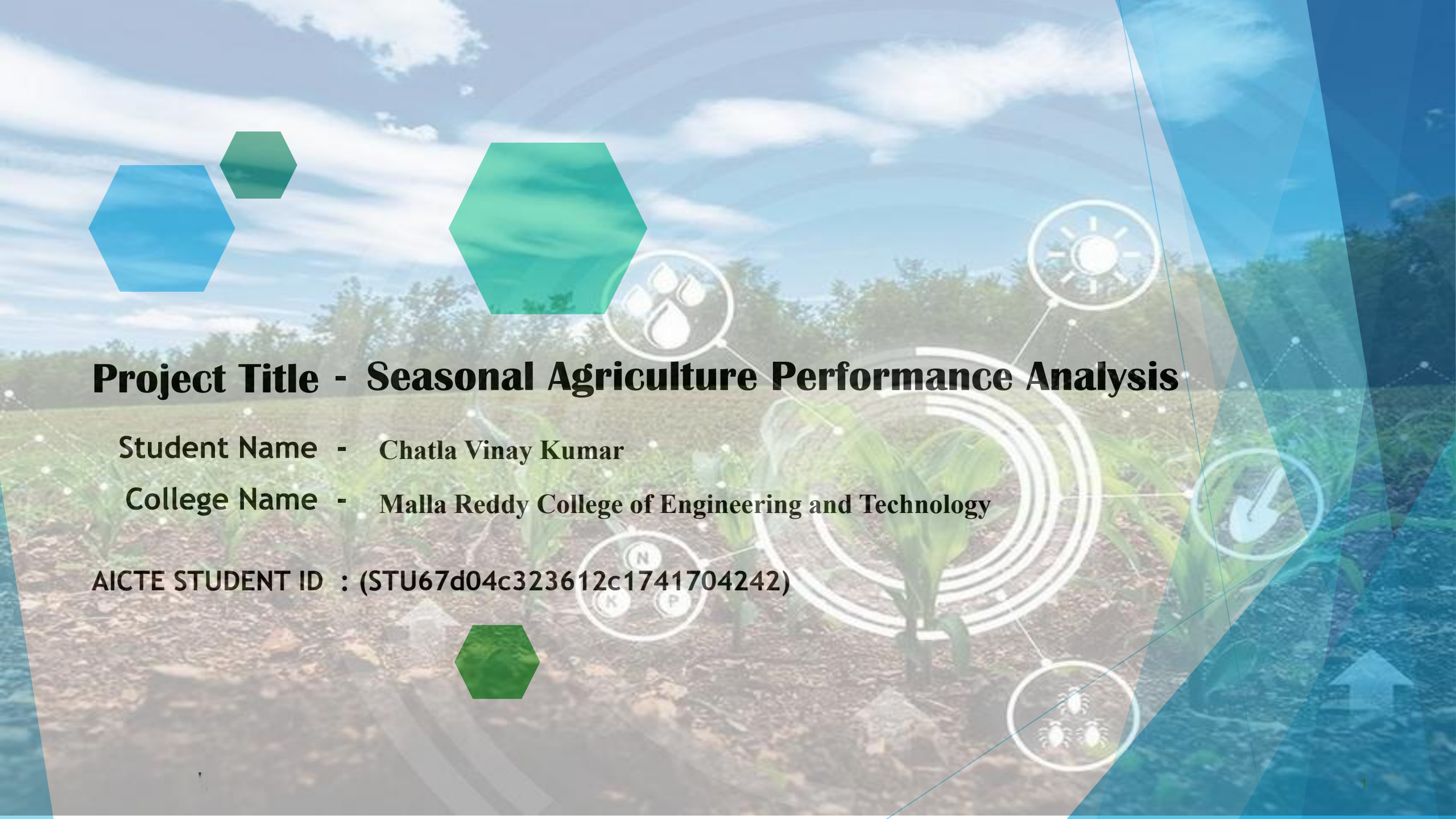




Project Title - Seasonal Agriculture Performance Analysis

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College Name - Malla Reddy College of Engineering and Technology

AICTE STUDENT ID : (STU67d04c323612c1741704242)

PROBLEM STATEMENT

- Agricultural activities are influenced by seasonal variations in environmental conditions, farming practices, resource availability and market conditions. As a result, agricultural performance may differ from one season to another. However, raw agricultural data does not clearly explain how agricultural performance changes across seasons or what patterns can be observed in different seasonal conditions. The problem is to analyze the given agricultural dataset and investigate seasonal differences in agricultural performance by identifying meaningful patterns, trends, relationships and variations within the available data.



Project Description

The primary aim is to investigate how agricultural outcomes vary across different seasons, uncover meaningful patterns, and provide evidence-based recommendations for improved planning and resource management. By analyzing a dataset containing environmental, agronomic, and economic variables, the project demonstrates how data science can support sustainable agriculture.

The dataset includes diverse features such as crop yield, rainfall, temperature, humidity, soil properties, fertilizer and pesticide usage, irrigation methods, water consumption, production costs, revenue, and profit. These variables provide a holistic view of agricultural activities across Kharif, Rabi, and Summer seasons. The analysis begins with data exploration and cleaning, addressing missing values, standardizing categories, and preparing the dataset for reliable insights.

Subsequent steps involve seasonal comparisons to evaluate yield, profitability, and resource usage. Statistical methods such as correlation analysis are applied to identify significant differences and relationships between environmental conditions and agricultural outcomes. Visualizations including boxplots, scatterplots, bar charts, and heatmaps highlight seasonal trends, anomalies, and variations across districts and states.

Key analytical questions include:

How do yield and profitability vary across seasons?

What environmental factors most strongly influence outcomes?

Are resource usage patterns consistent across regions?

Which crops or districts demonstrate resilience across multiple seasons?

What unusual or unexpected patterns emerge?

The project concludes with insight generation and recommendations, such as identifying the most profitable season, optimizing irrigation efficiency, improving fertilizer usage, and promoting crop diversification. These findings provide actionable guidance for farmers, policymakers, and researchers, supporting better seasonal agricultural planning and long-term sustainability.

WHO ARE THE END USERS?

- 1. Farmers and Growers
- 2. Agricultural Researchers and Analysts
- 3. Agri-businesses and Supply Chain Managers
- 4. Policy Makers and Government Agencies
- 5. Students and Educators
- 6. Environmental and Sustainability Experts



Technology Used



Python – Core programming language for data analysis and visualization.

Jupyter Notebook – Interactive environment for running code, documenting analysis, and presenting results.

Pandas – Data manipulation and cleaning (handling missing values, grouping, aggregation).

NumPy – Numerical computations and efficient array operations.

Matplotlib – Static visualizations (histograms, bar charts, scatter plots).

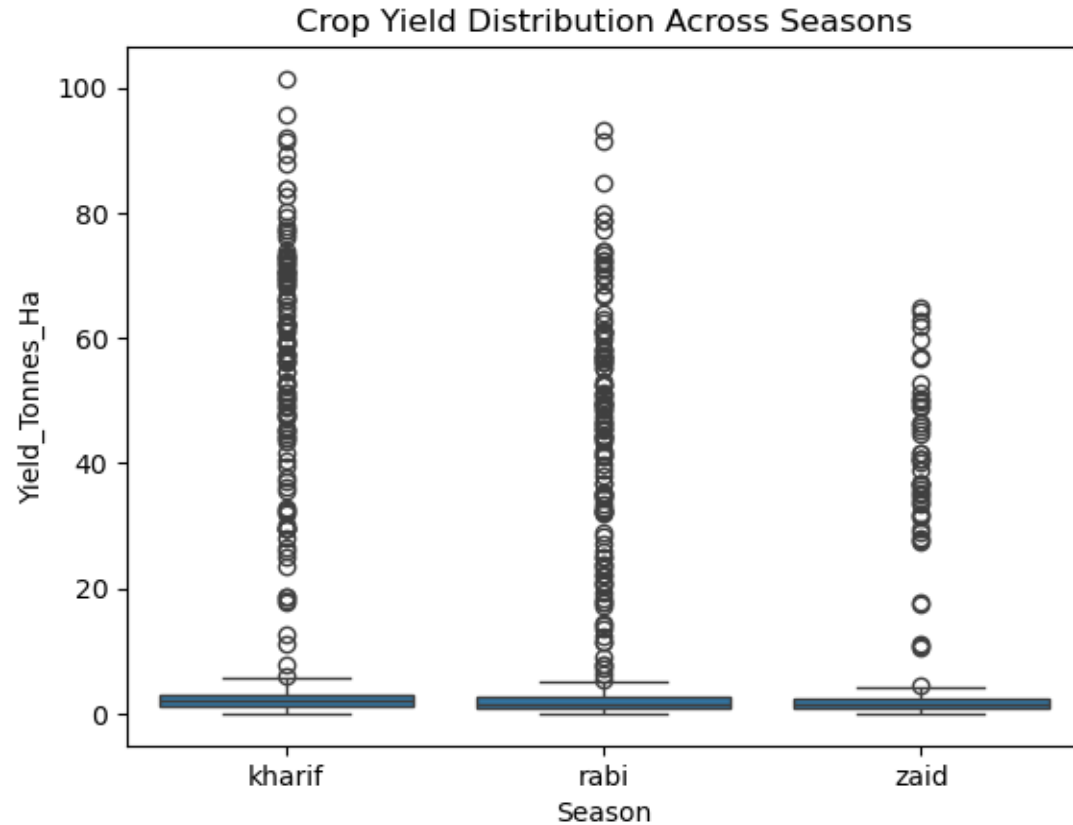
Seaborn – Advanced statistical visualizations (boxplots, heatmaps, regression plots).

Plotly – Interactive charts and dashboards for deeper exploration.

SciPy / Statsmodels – Statistical testing (ANOVA, correlation analysis).

Excel – For dashboarding and reporting outside Python.

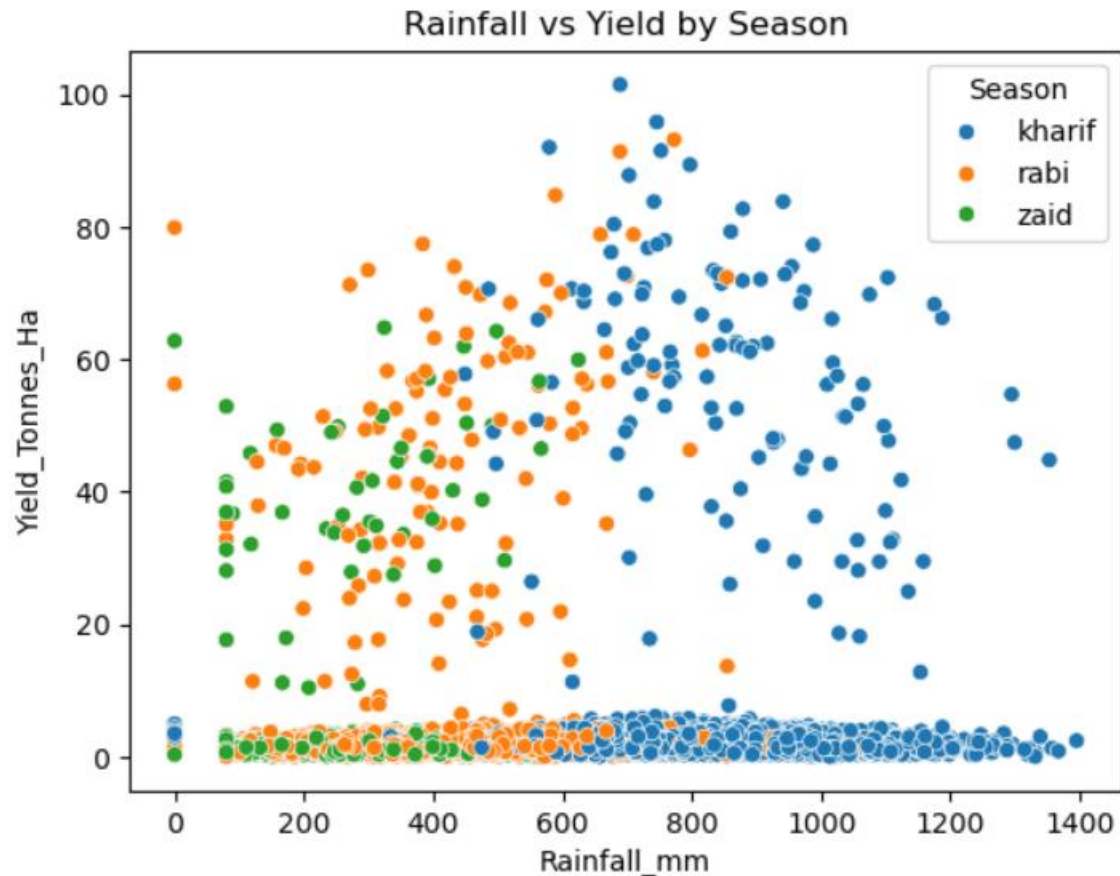
RESULTS



- The chart explains that seasonal agricultural performance is shaped by water availability and climate conditions. Rabi is the most stable and profitable season, Kharif is highly variable due to rainfall dependency, and Zaid is generally low-yield but shows potential through innovative practices. These insights emphasize the importance of season-specific planning, resource management, and climate adaptation strategies.

RESULTS

```
sns.scatterplot(x="Rainfall_mm", y="Yield_Tonnes_Ha", hue="Season", data=df)
plt.title("Rainfall vs Yield by Season")
plt.show()
```



The chart explains how rainfall correlates with yield differently across seasons:

Kharif Season :

Yields are strongly influenced by rainfall.

Higher rainfall generally corresponds to higher yields, but variability is noticeable.

This reflects the dependence of Kharif crops on monsoon rains.

Rabi Season :

Rainfall has less impact on yield because irrigation is more commonly used.

Yields are relatively stable across different rainfall levels.

This season shows resilience against rainfall fluctuations.

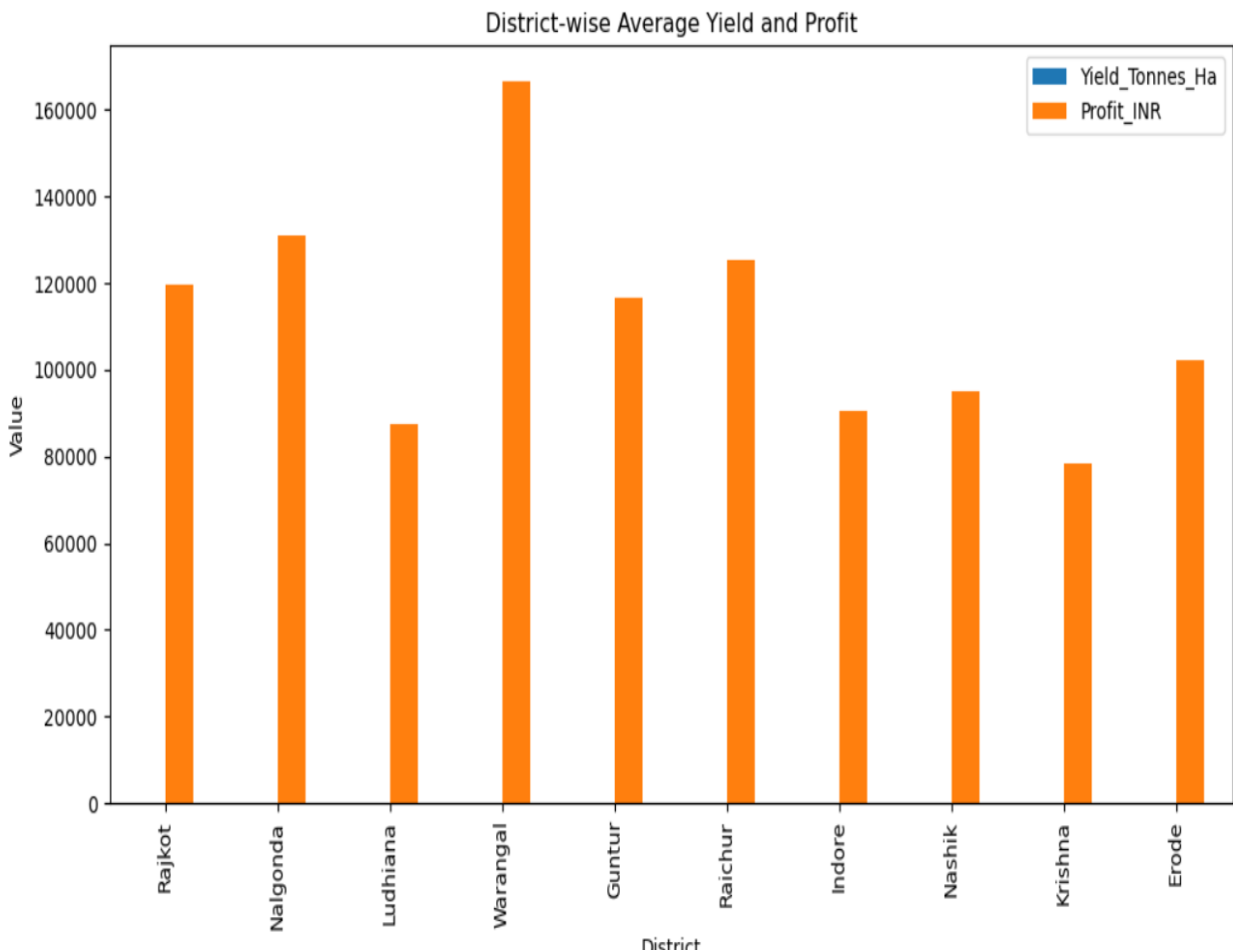
Zaid Season :

Rainfall is minimal, and yields are generally lower.

Outliers suggest that some farms achieve good yields despite limited rainfall, likely due to effective irrigation or crop choice.

RESULTS

District	Yield_Tonnes_Ha	Profit_INR
Rajkot	6.015810	119463.276808
Nalgonda	5.788602	131125.836145
Ludhiana	5.779508	87590.831382
Warangal	5.482773	166449.962085
Guntur	5.458974	116411.417949



This chart highlights that high yield does not always mean high profit. Districts like Warangal prove that profitability can be driven by factors beyond yield, such as crop choice, market access, or cost management. Conversely, districts with high yields but lower profits may need to optimize resource usage or improve market strategies.

Rajkot has the highest average yield (~6 tonnes/ha) but moderate profit compared to Warangal.

Nalgonda shows slightly lower yield than Rajkot but higher average profit, suggesting better economic efficiency.

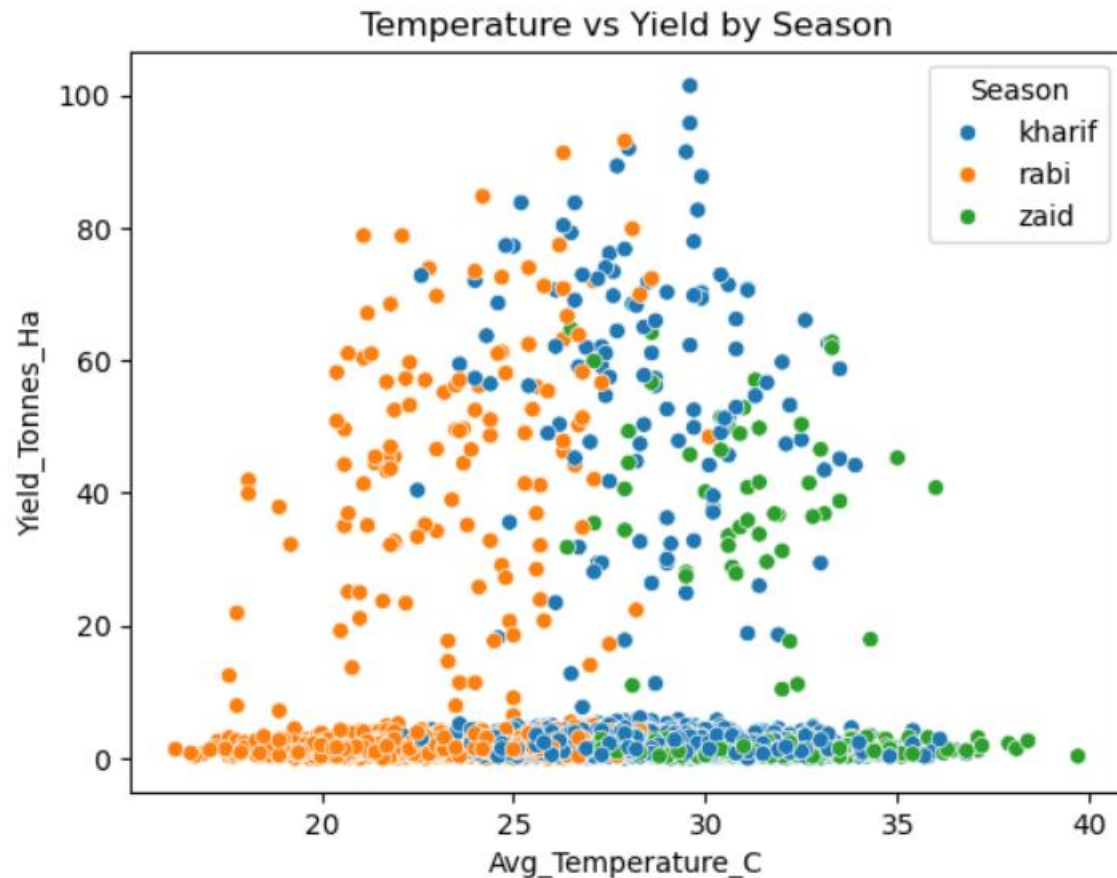
Ludhiana has strong yield but relatively lower profit, indicating possible higher costs or market differences.

Warangal demonstrates moderate yield but the highest profit among the top districts, showing strong economic returns despite lower productivity.

Guntur maintains balanced performance with good yield and profit.

RESULTS

```
sns.scatterplot(x="Avg_Temperature_C", y="Yield_Tonnes_Ha", hue="Season", data=df)
plt.title("Temperature vs Yield by Season")
plt.show()
```



The chart explains that temperature influences yield differently depending on the season:

Kharif :

Yield tends to fluctuate with temperature.

Crops are sensitive to both rainfall and temperature, so extreme heat or cooler monsoon conditions can reduce yield.

Rabi :

Yields are relatively stable across temperature ranges. Since irrigation is common, temperature has less direct impact compared to rainfall.

Moderate temperatures favor consistent productivity.

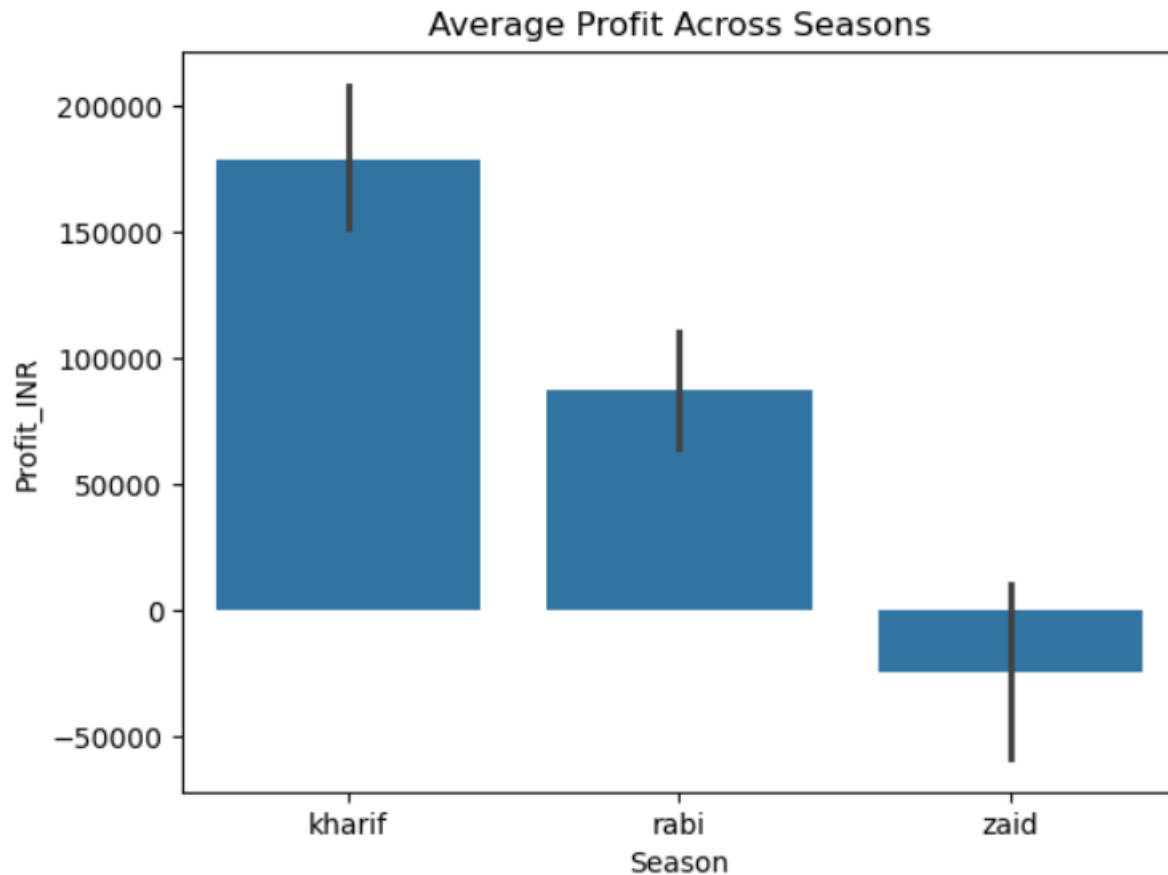
Zaid :

Yields are generally lower at higher temperatures due to heat stress and limited water.

A few outliers show good yields even in hot conditions, likely due to effective irrigation or heat-resistant crop varieties.

RESULTS

```
sns.barplot(x="Season", y="Profit_INR", data=df, estimator=np.mean)
plt.title("Average Profit Across Seasons")
plt.show()
```



The chart explains that profitability varies significantly by season:

Kharif Season (Monsoon):

Shows the highest average profit (~175,000 INR).

This reflects strong yields when rainfall is favorable, making it the most profitable season overall.

Rabi Season (Winter):

Average profit is moderate (~90,000 INR). Reliance on irrigation stabilizes yields, but profitability is lower compared to Kharif.

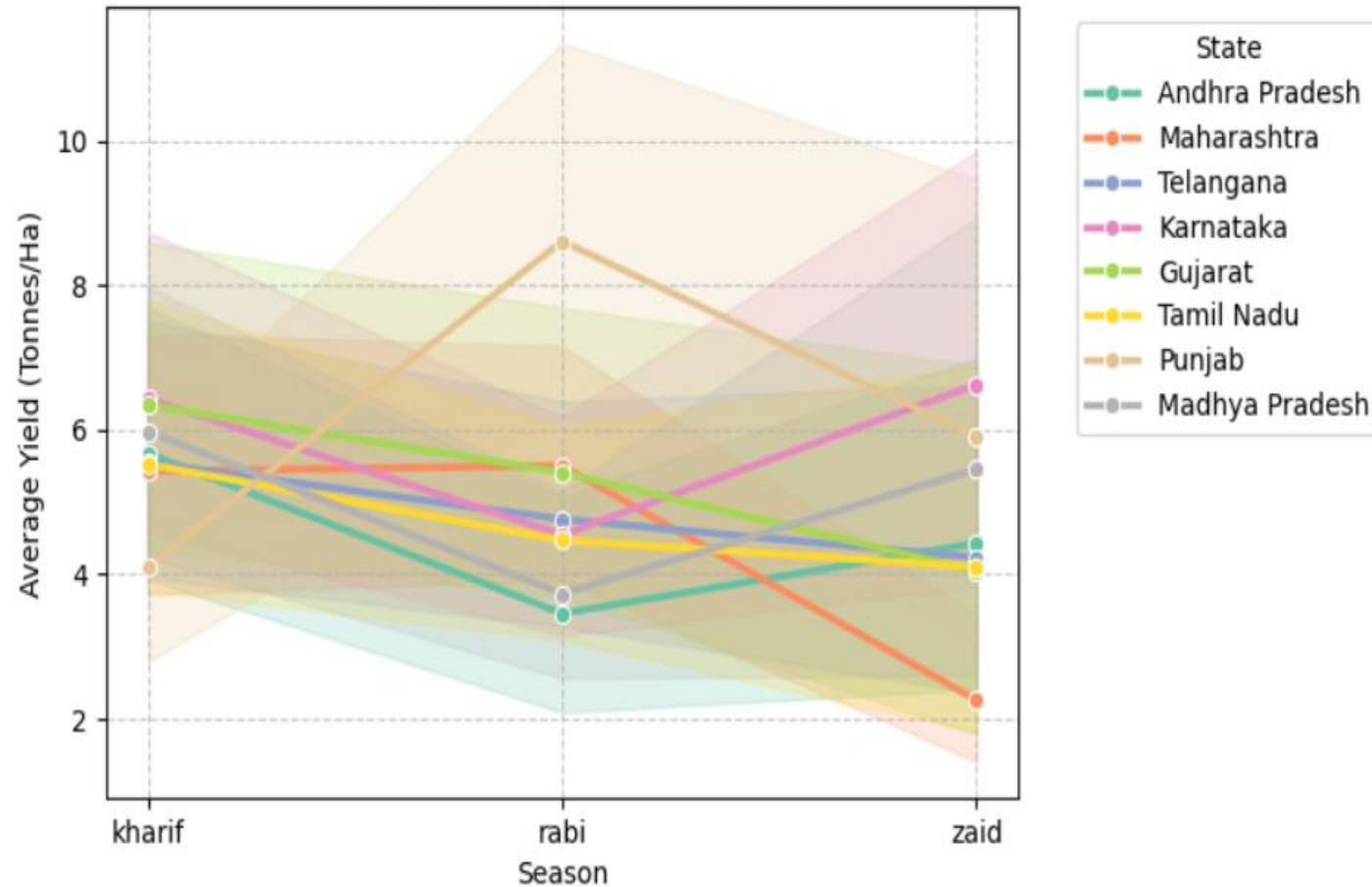
Zaid Season (Summer):

Average profit is lowest, slightly below zero.

Harsh summer conditions and limited water availability make this season economically risky.

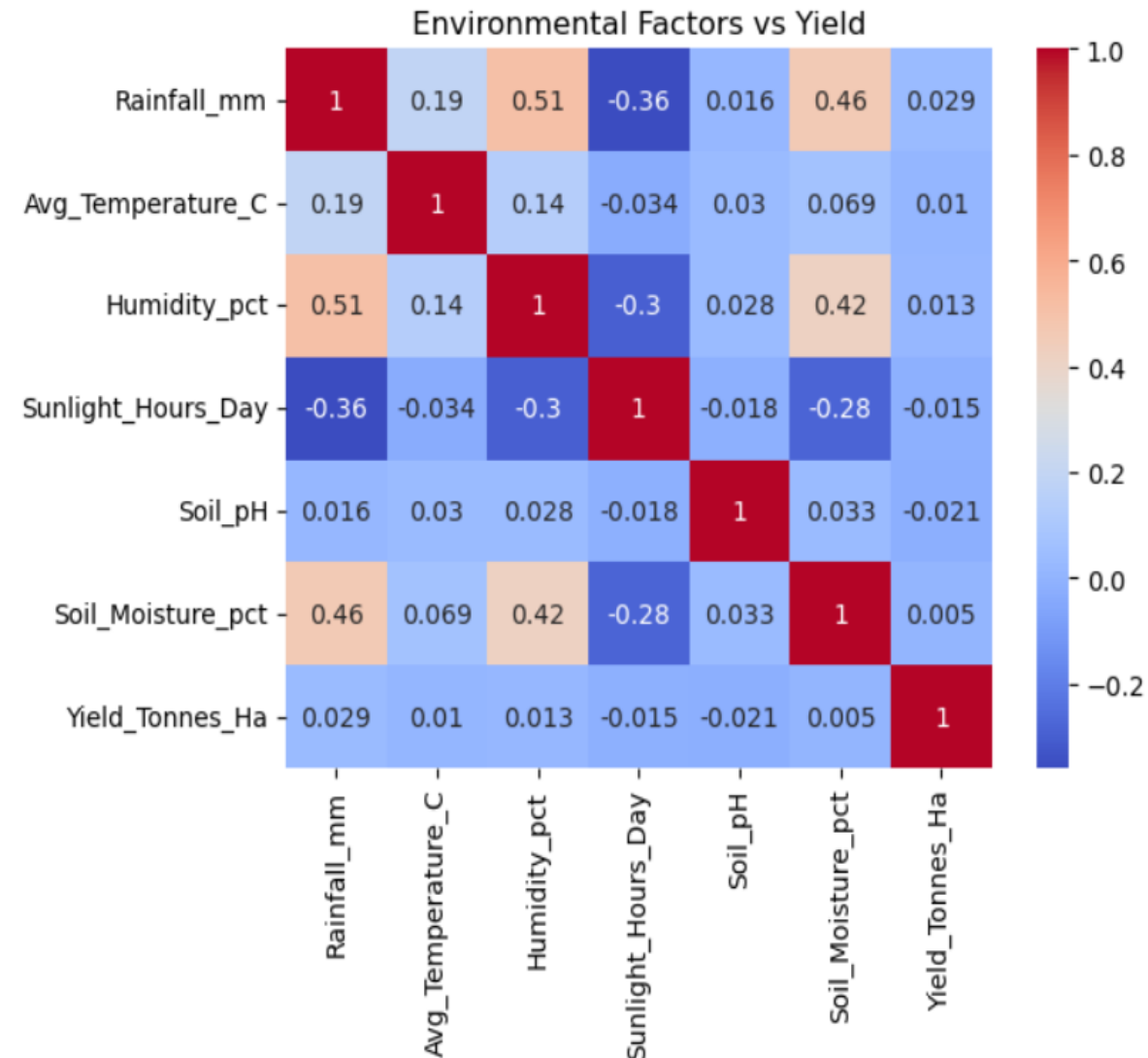
RESULTS

Yield Patterns Across States



The chart explains that agricultural productivity is highly seasonal and region-specific. While Rabi often provides stable and high yields in irrigated states, Kharif depends on monsoon rainfall, and Zaid is generally the weakest season due to harsh summer conditions. This visualization helps identify which states are resilient and which need targeted interventions for specific seasons.

RESULTS



The heatmap explains that water availability (rainfall, soil moisture) is the strongest driver of yield, while temperature and humidity play secondary roles. Sunlight helps, but excessive heat can be harmful. Soil pH has minimal impact within normal ranges.

Rainfall vs Yield:

Strong positive correlation. More rainfall generally improves yield, especially in rain-fed seasons like Kharif.

Soil Moisture vs Yield:

Positive correlation, showing that adequate soil water content supports higher productivity.

Sunlight Hours vs Yield:

Moderate positive correlation, indicating that longer sunlight exposure helps crop growth, though excessive heat may reduce efficiency.

Temperature vs Yield:

Often shows a weaker or negative correlation. Very high temperatures (especially in Zaid) reduce yields due to heat stress.

Humidity vs Yield:

Mixed correlation. Moderate humidity supports growth, but excessive humidity can cause diseases, lowering yield.

Soil pH vs Yield:

Correlation close to zero or weak. This suggests yield is less sensitive to pH within the dataset's range, though extreme pH values outside the range could still matter.

Future scope

Looking ahead, this project has the potential to grow into something far more impactful. While the current analysis provides valuable insights into seasonal yields, profits, and environmental factors, there are several directions in which it can be expanded:

Predictive Modeling: Moving beyond descriptive analysis, the next step could be building machine learning models that forecast yields and profits based on rainfall, temperature, and soil conditions. This would help farmers plan ahead with greater confidence.

Real-time Data Integration: Incorporating live weather feeds and IoT sensor data (soil moisture, pH, humidity) would make the system dynamic, allowing for timely recommendations and alerts during critical farming stages.

Interactive Dashboards: Developing user-friendly dashboards with tools like Power BI or Plotly would make the insights more accessible to policymakers, researchers, and farmers, enabling them to explore district-wise and season-wise trends interactively.

Decision Support System: A recommendation engine could be built to suggest optimal crop choices, irrigation methods, and fertilizer usage tailored to each district and season. This would transform the project into a practical decision-making tool.

Policy and Economic Analysis: Extending the scope to include cost-benefit studies and subsidy impact assessments would help governments design better agricultural policies and support schemes.

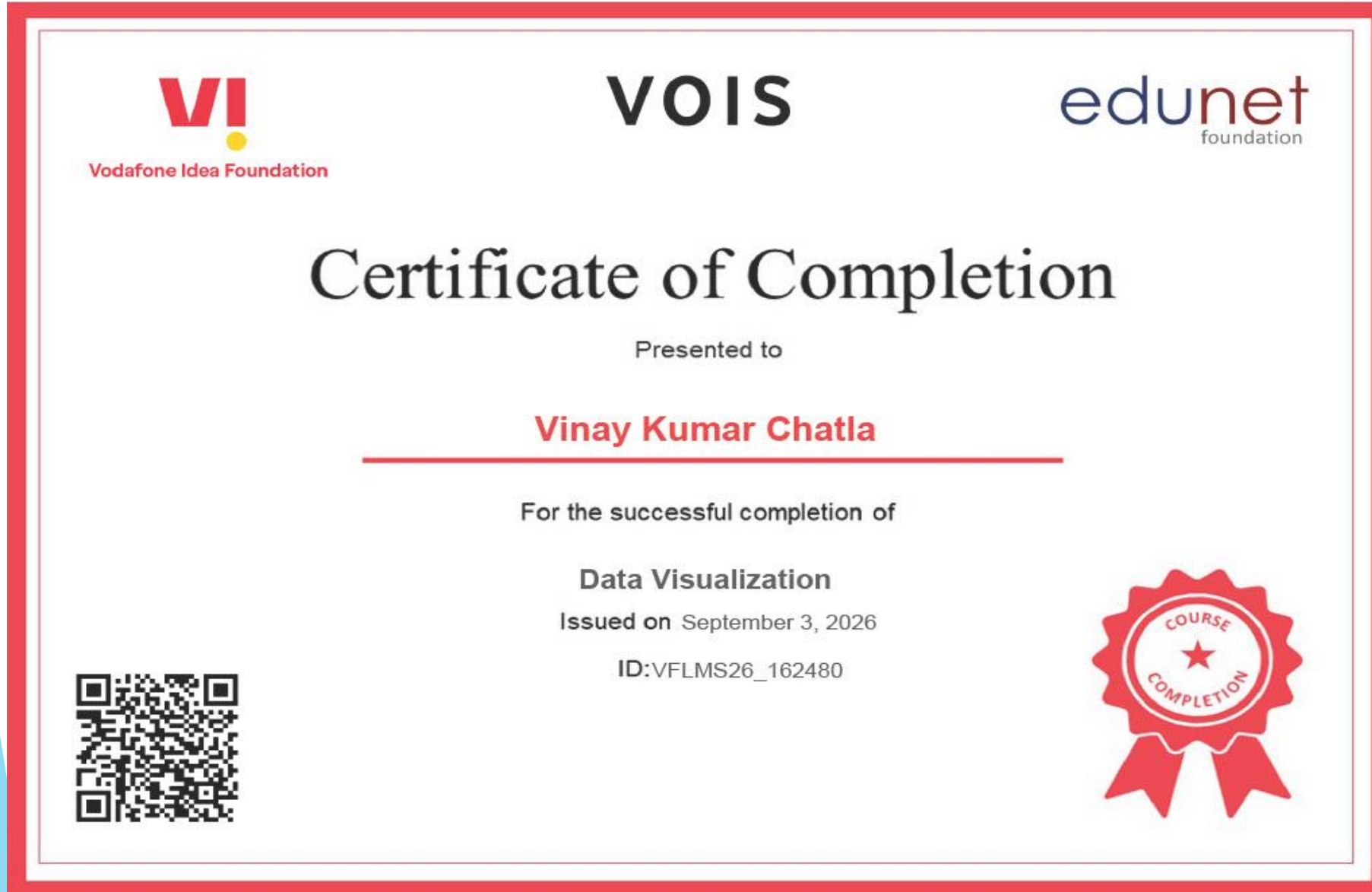
Scalability: Finally, the framework can be scaled to cover multiple states or even countries, enabling comparative studies and broader agricultural planning at a national or global level.

GitHub Link



<https://github.com/vinay64414/Seasonal-Agriculture-Performance-Analysis-.git>

VOIS Course completion certificate - Course name- Data Visualization





THANK YOU!